Rishav Dutta

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🌐 **Digital Portfolio:** https://rishav0123.github.io/portfolio/ **Total Experience: 4.5 Years**

PROFESSIONAL SUMMARY

Senior Data Engineer and Cloud Automation Specialist with 4.5+ years of enterprise experience and extensive hands-on work building end-to-end data products and AI-powered platforms through personal and open-source initiatives. Expert in Python-driven automation, serverless AWS architectures, distributed data processing, and real-time analytics, with experience owning systems from design to deployment and observability. Combines strong engineering fundamentals with a product-focused mindset, delivering scalable, secure, and cost-optimized data solutions across both enterprise and independent development environments.

KEY SKILLS

- **Languages:** Python, SQL, JavaScript, Java, C#
- **Data Engineering & Big Data:** PySpark, Apache Kafka, Airflow, AWS Glue, Spark SQL, ETL/ELT, Batch & Streaming Pipelines
- **Databases:** Amazon Redshift, PostgreSQL, MySQL, MongoDB, Teradata, DynamoDB
- **Cloud & DevOps:** AWS (Lambda, EC2, S3, Step Functions, CloudTrail, IAM, EventBridge, CloudWatch), Docker
- **Frameworks & Tools:** Django, Playwright, Pandas, NumPy, Jupyter, Git, REST APIs
- **BI & Visualization:** Power BI, Tableau
- **Other Tools:** Supabase, Frappe, ERPNext, WordPress Plugin Development
- **Soft Skills:** Leadership, Communication, Critical Thinking, Agile Methodologies

PROFESSIONAL EXPERIENCE

Tata Consultancy Services (TCS) – ITPL, Bangalore
Data Engineer / Python Developer Feb 2024 – Present

- Automated Tableau report retrieval using REST API and scheduled email delivery.
- Built server health check scripts for storage monitoring and auto-cleanups.
- Migrated legacy reporting to Power BI, enhancing visualization and accessibility.

Integrated automated workflows into UAC for report generation, server checks, and data refreshes.
Technologies: Tableau, Power BI, UAC, Python (smtplib, os, shutil, psutil)

Tata Consultancy Services (TCS) – ITPL, Bangalore
Data Engineer July 2021 – Nov 2022

- Transformed automotive and sales data using SQL, Redshift, and MS SQL Server.
- Implemented Price Waterfall Model for pricing logic and discounts.
- Optimized queries with CTEs, indexing, and stored procedures.

- Built role-based workflow authorization and platform logic with reusable modules.
  Technologies: Amazon Redshift, SQL Server, Stored Procedures, CTEs

Tata Consultancy Services (TCS) – ITPL, Bangalore
AWS Python Automation Specialist Dec 2022 – Feb 2024

- Built Lambda functions to manage image lifecycle events (format validation, archival).
- Automated EC2 resource scheduling, saving 35% compute cost.
- Used CloudTrail, IAM, and EventBridge for secure, event-driven automation.
- Orchestrated workflows via AWS Step Functions for process control.

Technologies: Lambda, EC2, S3, S3 Glacier, CloudTrail, EventBridge, CloudWatch, Step Functions, Python, Boto3, PIL

PROJECTS

Sentimatix — AI-Powered Financial Intelligence Platform

- GitHub: https://github.com/Rishav0123/sentimatix.git
- Site: https://sentimatix.onrender.com/
- Developed an end-to-end financial analytics platform connecting React/TypeScript frontend with FastAPI backend.
- Implemented Model Context Protocol (MCP) for agentic workflows orchestrating RAG searches, real-time news scraping, and LLM-based sentiment analysis.
- Deployed on Render with Dockerized CI/CD pipeline, solving dependency challenges for scalable insights.
- Built scalable ingestion pipeline using PySpark and Supabase (pgvector) for semantic search.
- Delivered responsive dashboard with React 18 and Tailwind CSS, supporting real-time WebSocket updates.

Event Management System

- Designed backend architecture using Supabase for database and authentication.
- Developed Edge Functions for real-time triggers, user actions, and API requests.
- Built REST APIs for event creation, registration, and notifications with role-based access control.
- Tech Stack: Supabase, JavaScript, PostgreSQL, Edge Functions, REST APIs

EDUCATION

Bachelor of Technology (B.Tech/B.E.) National Institute of Technology, Durgapur (2017 – 2021)
Majors: Metallurgical and Materials Engineering

CERTIFICATIONS

- AWS Certified Data Engineer – Associate (DEA-C01)
- Big Data Processing with PySpark — Coursera

PERSONAL ATTRIBUTES

Persistent, detail-oriented, and entrepreneurial. Blend technical execution with product thinking, always seeking best practices and scalable solutions. Thrive on collaborative problem-solving and future-proofing data systems.